POCKET AI ASSISTANT · ENCLOSURE, FIRST STEP

The cardboard mock-up

The electronics work. The device speaks at volume cap 45. What is left is giving it a body, and every enclosure starts the same way: the real parts, arranged at full size until everything fits, the plug reaches, and sound gets in and out. You own spares of every module, so you can lay them out unpowered and leave the working prototype wired. No soldering, no scope, no risk.

Where you are

1–2 · done	3 · done	4a · today	4b–4e · next
Prototype + cloud	**It speaks**	**Mock-up**	**Case**
USB prototype, Wi-Fi, assistant replies	amplifier, cap 45, GAIN 15 dB	spare parts, cardboard box, test fit	paint test, frame design, brass, permanent wiring

0:00 Close out Step 11

- [ ] **PASS 11C · SPEAKS AT CAP 45**
  Five replies in a row are clearly audible, with no reset, no harsh buzz, and nothing warm. The firmware's peak monitor stays under 2.53 V. Tell Claude when this is true so it can write the record.

0:05 Confirm the last measurements

Six numbers are still open. Each has an estimate you can use as it is; confirm it against the real part and type a number only where yours differs. Plain millimetres; the page remembers them in this browser.

MEASUREMENT	ESTIMATE	CONFIRM WITH THE REAL PART	MM
OLED lit area length	21.7	Working A1 on; measure only the part that glows, long side	
OLED lit area width	10.9	The glowing part, short side. Also mark *where* it sits on the board; it is usually not centred	
USB socket offset	1.7	Board edge to socket face; your caliper reading from 28 September	
USB plug width	12.35	Your cable's plastic plug body, side to side. The estimate is the largest the USB-C standard allows	
USB plug height	6.5	The same plug body, top to bottom; also the standard's largest	
Bend room	≈ 35	Plug the cable into a spare board, bend it 90°, measure board edge to the outside of the cable (30–40 is typical)	

The plug estimates are worst case on purpose: an opening sized for the largest plug fits any cable.

0:15 Gather the spare parts

Use one spare of each module as the stand-in for the final device. The working prototype stays on the breadboard, untouched. Counts are from `docs/INVENTORY.md`.

PART	MEASURED SIZE	OWNED	IN THE PROTOTYPE	SPARES
ESP32-C3 SuperMini	22 × 18	10	1 (A1)	9
OLED screen	27 × 27	5	1	4
INMP441 microphone	Ø13	5	1	4
MAX98357 amplifier	17 × 17	3	1	2
Speaker	20 × 13	6	1	5
Button	5 × 5	420	1	plenty

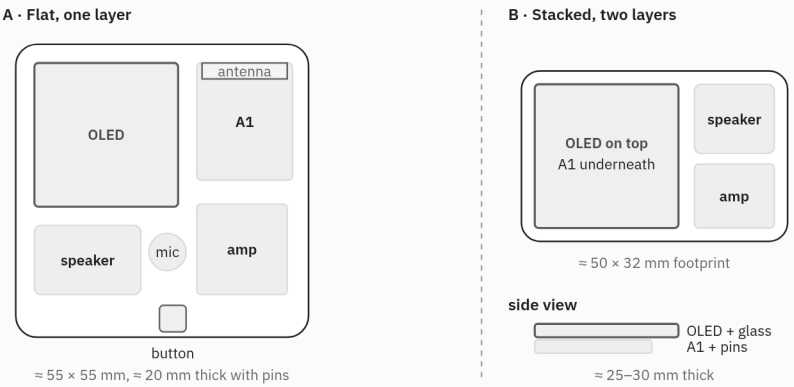
BEFORE YOU LAY THEM OUT

- **Keep the spares unpowered.** No USB, no battery; they sit on cardboard, so nothing can short.
- **Check for header pins.** A spare with pins soldered on sits taller than a final part wired with short leads; one without pins is close to final thickness. Note which you used.
- **The screen window comes from the working A1.** An unpowered spare shows no lit area, so switch on the working screen to see where the glow sits, and mark that position on the spare's footprint.
- **Leave wire room.** The spares are not wired, so keep 2–3 mm around each one for the wires that will join them.

The paper templates are still in `edu/pages/mockup-templates-letter.pdf` if a spare is missing. Page 2 of that file has the box nets you will use below. Print at **Actual size** and check that the ruler measures 100 mm.

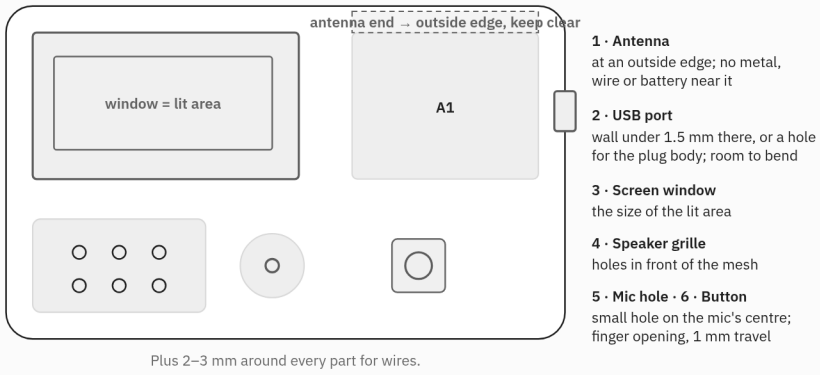
0:30 Try two arrangements

Lay the spare parts on a sheet of cardboard. There is no single right answer; try both of these and keep the one that is smaller and still obeys every rule in the next section.



Flat is thinner; stacked is smaller across. Sizes are rough first guesses from your measurements; the real parts tell the truth. In both, the antenna end of A1 sits against an outside edge.

0:45 The rules every layout must pass



Most first layouts fail on the USB port or the antenna. The socket sticks out 1.7 mm and the plug's plastic starts 2 mm further, so a thick wall there stops the cable reaching. The SuperMini already needed a transmit-power cap to be heard at all, so its antenna end needs open air.

1:00 Build the cardboard box

1. **Start from a box net.** Page 2 of `edu/pages/mockup-templates-letter.pdf` has a fold-up box and lid for each arrangement. If your real layout is bigger, trace its outline with 2–3 mm margin instead and cut it twice: floor and lid.

2. **Set the wall height** to the tallest spare as it lies, plus 3 mm: about 21 mm if the amplifier keeps its pins, about 12 mm if you plan to trim them. Measure the real stack; for the stacked layout that is the screen on top of A1.

3. **Fold or tape the walls** up, set the spares inside, and mark the openings where the parts actually sit: USB, screen window, speaker grille, mic hole, button.

4. **Cut the openings** with a craft knife on a cutting mat.

1:20 Test fit with the real board

1. **Plug the USB cable into the spare board inside the box** through the wall opening: does the plug reach the socket, and can the cable bend away? The spare has no power, so this is safe.
2. **Press a finger** through the button opening: could it reach a button there?
3. **Check the window** against the working A1's lit screen: hold it under the window and see whether the glow fills the opening.
4. **Photograph** the box from above and from each side, with a ruler in the frame, and write down its outside size.

- [ ] **MOCK-01 · THE LAYOUT FITS**
  Every part fits with margin, the antenna end is at an outside edge, the USB plug reaches, and all five openings line up. Send Claude the outside size (L × W × H in mm), which arrangement you chose, and whether the spares had header pins.

After the mock-up

NEXT	WHAT IT INVOLVES	NEEDS
Paint test coupons	sand brass offcuts, wipe with 91 % alcohol, spray the two whites, test adhesion after curing	owned paint and sandpaper
Frame design	Claude turns your mock-up size and the measurements into a frame drawing, replacing the old CAD, which models a withdrawn design	MOCK-01
Cut the brass	only after the frame drawing is checked against the mock-up	frame design
Permanent wiring	replace breadboard jumpers with short soldered wires, one module at a time, testing after each	the final layout
Sleep button (optional)	a button on GPIO0 that puts A1 to sleep and wakes it; Claude writes the firmware	one owned button

Sources. Part sizes are the builder's caliper measurements of 28 September; the speaker follows its datasheet; spare counts come from `docs/INVENTORY.md`. The OLED lit area is a typical value for a 0.96 inch 128 × 64 module, and the USB plug estimates are the USB-C standard's largest plug body; both stand until you measure. Fit guidance from `plan/concepts/14-fit-and-radio.md`; `cad/README.md` marks the old CAD as stale and asks for exactly this 1:1 model. If this page and the repository differ, the repository wins.

What is and is not verified. The layout sizes are first estimates; spare parts and a cardboard box check space only, not radio range. A spare may differ slightly from the parts in the working prototype. The checkboxes and table values are remembered only in this browser.